

EXERCISE

13C

Theoretical and experimental probability

INDIVIDUAL
PATHWAYSeBook *plus*

Activity 13-C-1

Snakes and
ladders 1
doc-4167

Activity 13-C-2

Snakes and
ladders 2
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Activity 13-C-3

Snakes and
ladders 3
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FLUENCY

- 1 **WE7** A random card is drawn from a shuffled pack of 52 cards. Find the probability of selecting:

- a** an ace
c a heart

- b** a black card
d a 7.

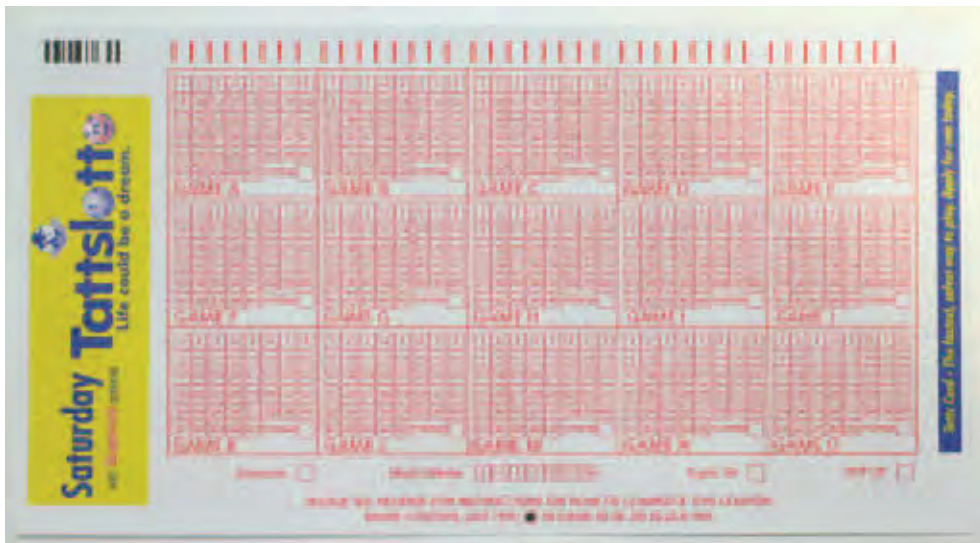


- 2 A pencil case contains 4 blue pens, 7 black pens and 9 red pens. If a pen is chosen, at random, from the pencil case, find:

- a** $P(\text{red pen})$ **b** $P(\text{black pen})$ **c** $P(\text{not a blue pen})$.



- 3 A coin is tossed and a die is rolled simultaneously.
- Use a two-way table to list the sample space.
 - What is the probability of getting a Head on the coin and an even number on the die?
- 4 **MC** A random number is picked from $N = \{1, 2, 3, 4 \dots 100\}$. What is the probability of picking a number that is a multiple of 2?
- A $\frac{49}{100}$ B $\frac{51}{100}$ C $\frac{50}{100}$ D $\frac{50}{101}$ E $\frac{49}{50}$
- 5 A green die and a blue die are rolled together. The number on each die is added to obtain the outcome.
- Display all possible outcomes graphically. How many outcomes are there?
 - What is the most probable outcome?
 - What is the least probable outcome?
 - What is the probability of an outcome being less than 5?
 - What is the probability that the outcome is 6?
- 6 **MC** Eight Nissans, six Toyotas, four Holdens and two Mitsubishi's enter a car rally. Assuming that all cars have equal chances of winning, what is the probability that a Toyota wins the rally?
- A $\frac{4}{10}$ B $\frac{3}{20}$ C $\frac{1}{5}$ D $\frac{3}{10}$ E $\frac{1}{4}$
- 7 At a birthday party, some cans of soft drink were put in a container of ice. There were 16 cans of Coke, 20 cans of Sprite, 13 cans of Fanta, 8 cans of Sunkist and 15 cans of Pepsi.
- If a can was picked at random, what is the probability that it was:
- a can of Pepsi?
 - not a can of Fanta?
- 8 **MC** In Tattslotto, 6 numbers are drawn from the numbers 1, 2, 3, ... 45. The number of different combinations of 6 numbers is 8 145 060. If you buy 1 ticket, what is the probability that you will win the draw?



- A $\frac{1}{8\,145\,060}$ B $\frac{1}{45}$ C $\frac{45}{8\,145\,060}$ D $\frac{1}{6}$ E $\frac{6}{8\,145\,060}$
- 9 **WE8** If a fair coin is tossed 400 times, how many tails are expected?
- 10 If a fair die is rolled 120 times, how many threes are expected?
- 11 **WE9 MC** A survey of high school students asked 'Should Saturday be a normal school day?' 350 students voted yes, and 450 voted no. What is the probability that a student chosen at random said no?
- A $\frac{7}{16}$ B $\frac{9}{16}$ C $\frac{7}{9}$ D $\frac{9}{14}$ E $\frac{1}{350}$